

Exact Differential Equation - Step-by-Step Solution

Problem: Solve

$$2xy \, dx + (x^2 - 1) \, dy = 0$$

Step 1: Identify $M(x,y)$ and $N(x,y)$

The standard form of an exact differential equation is:

$$M(x,y) \, dx + N(x,y) \, dy = 0$$

Comparing with $2xy \, dx + (x^2 - 1) \, dy = 0$, we get:

$$M(x,y) = 2xy \text{ and } N(x,y) = x^2 - 1$$

Step 2: Check whether the equation is exact

For an equation to be exact, the partial derivatives must satisfy:

$$M_y = N_x$$

Differentiate $M = 2xy$ with respect to y . Since x is treated as a constant:

$$M_y = d(2xy)/dy = 2x$$

Differentiate $N = x^2 - 1$ with respect to x :

$$N_x = d(x^2 - 1)/dx = 2x$$

Because $M_y = N_x = 2x$, the equation is **exact**.

Step 3: Find the potential function $F(x,y)$

For an exact equation, there is a function $F(x,y)$ such that $F_x = M$ and $F_y = N$.

$$F_x = 2xy$$

Integrate with respect to x . During this integration, y is treated as a constant:

$$F(x,y) = \int 2xy \, dx = y \int 2x \, dx = x^2 y + g(y)$$

The extra term $g(y)$ is included because the constant of integration may depend on y .

Step 4: Differentiate F with respect to y

$$F(x,y) = x^2 y + g(y)$$

$$F_y = x^2 + g'(y)$$

But F_y must equal $N(x,y) = x^2 - 1$. Therefore:

$$x^2 + g'(y) = x^2 - 1$$

$$g'(y) = -1$$

Step 5: Find $g(y)$

$$g(y) = \int (-1) \, dy = -y$$

Step 6: Write the implicit solution

Substitute $g(y) = -y$ into $F(x,y)$:

$$F(x,y) = x^2 y - y$$

For an exact equation, the solution is $F(x,y) = C$:

$$x^2 y - y = C$$

Factor out y :

$$y(x^2 - 1) = C$$

Equivalently, solving for y : $y = C / (x^2 - 1)$

Exact ODE pattern to remember

1. Identify M and N from $M dx + N dy = 0$.
2. Check $M_y = N_x$.
3. Integrate M with respect to x : $F = \int M dx + g(y)$.
4. Differentiate F with respect to y and compare with N to find $g(y)$.
5. Write the final implicit solution $F(x,y) = C$.